DESIGN COMPENDIUM

Analog and Mixed-Signal Circuit Design

“Designing CMOS analog and mixed-signal circuits with focus on performance, stability, and low-power optimization. ”

Nikhil Jindal

Mixed-Signal Engineering Overview

This compendium presents the design and simulation of CMOS analog and mixed-signal building blocks. Each project focuses on translating theoretical concepts into practical circuit implementations, with emphasis on performance metrics such as gain-bandwidth, power consumption, stability, and temperature behavior.

CORE COMPETENCIES FOCUSED IN THIS REPORT

Frequency Response Analysis	Operational Amplifier Stability	Active Filter Synthesis
Signal Conditioning & Filtering	Mixed-Mode Conversions	Feedback Control Systems

TABLE OF CONTENTS

*Note: All designs are verified via Cadence Virtuoso simulation

Single-Stage CMOS Op-Amp (Low-Power Design @ ±0.9V)

Overview

Designed and simulated a single-stage CMOS operational amplifier in Cadence Virtuoso, optimized for low-power operation while maintaining moderate voltage gain.

Design Targets

- Gain ≥ 50 dB
- Low power consumption (< 10 µW)
- Supply voltage: ±0.9 V
- Load capacitance: 10 pF
- Target GBW: ~5 MHz

Architecture

The amplifier is implemented using a single-stage differential topology consisting of:

- NMOS differential input pair
- PMOS current mirror active load
- Tail current source for biasing

This topology was selected to achieve high gain with minimal power consumption while keeping circuit complexity low.

Key Design Decisions

- Used PMOS current mirror load to increase output resistance and improve gain
- Chose low bias current (~2 µA) to minimize power consumption
- Operated input pair in moderate inversion (gm/Id based design) for gain–power tradeoff
- Maintained all transistors in saturation region to ensure proper amplification
- Load capacitance fixed at 10 pF, directly impacting bandwidth

Schematic

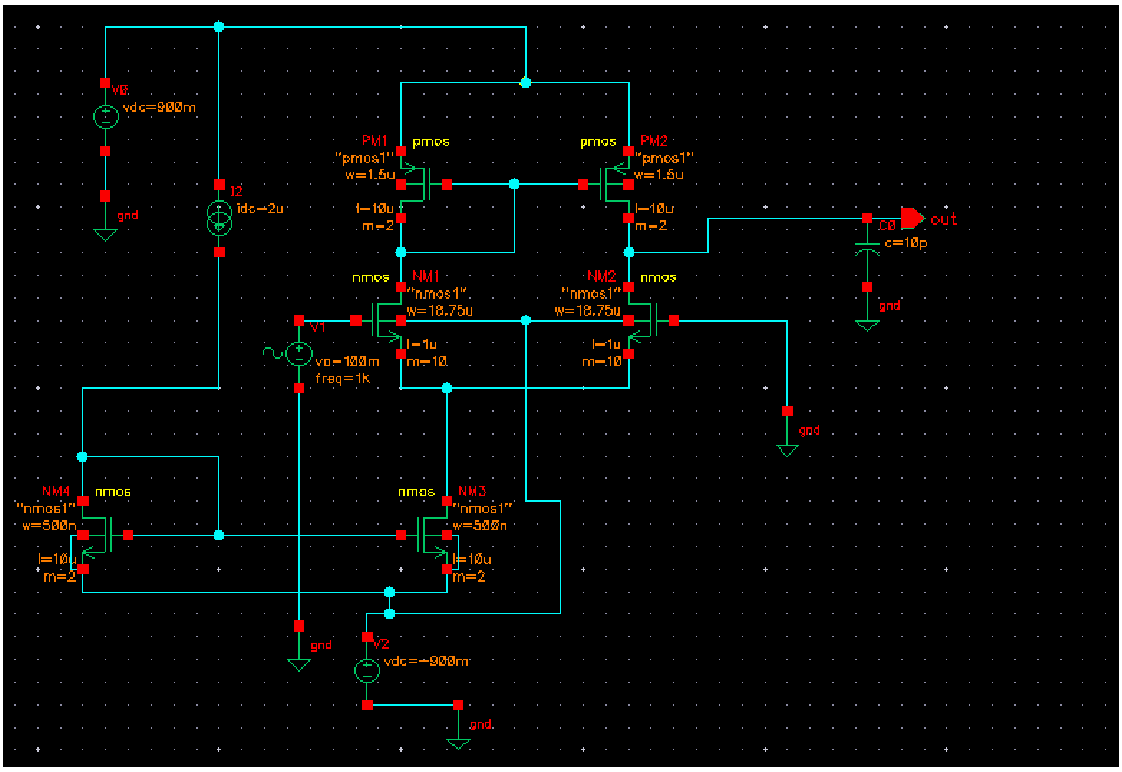

Fig 1:Schematic of Single Stage Op-Amp

Final Design Parameters

Parameter	Value
Supply Voltage	±0.9 V
Bias Current	2 µA
Load Capacitance	10 pF

Results

Parameter	Value
Gain	54.34 dB
Gain (linear)	~500
Bandwidth (-3 dB)	1.67 kHz
Power Consumption	5.15 µW
Input Noise (RMS)	3.60×10^{-12} V²
Slew Rate	0.2 V/µs

Key Plots

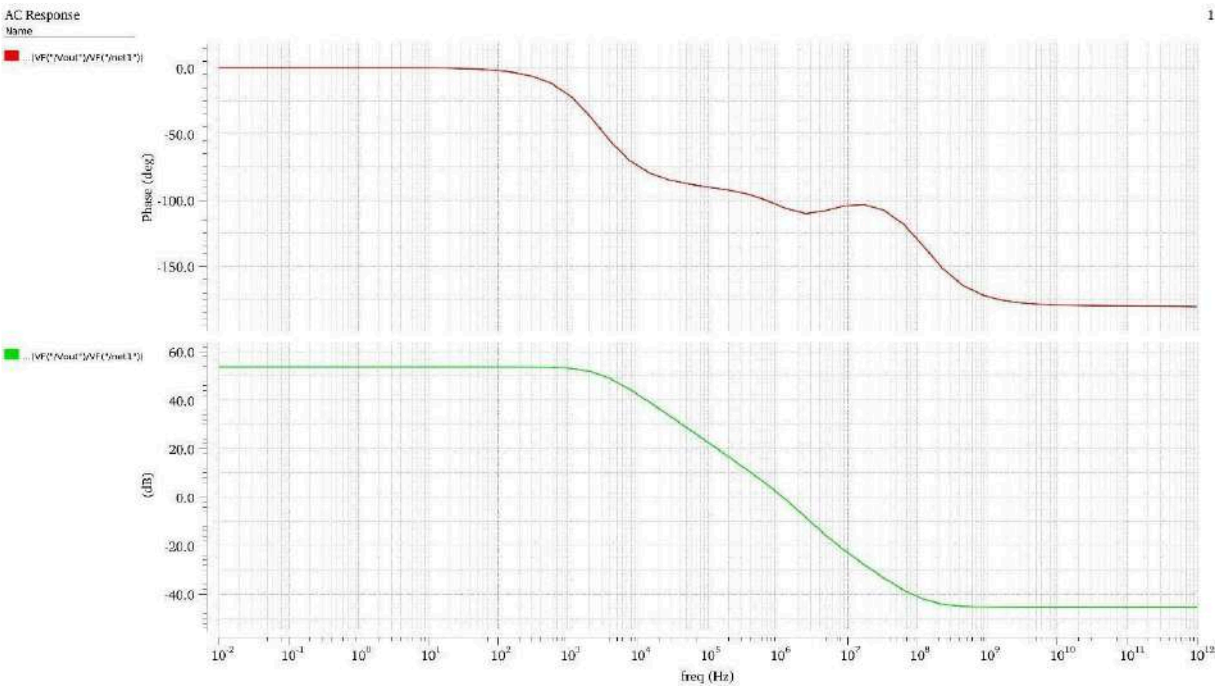

Fig 2: Gain and Phase Plot of Single Stage Op-Amp

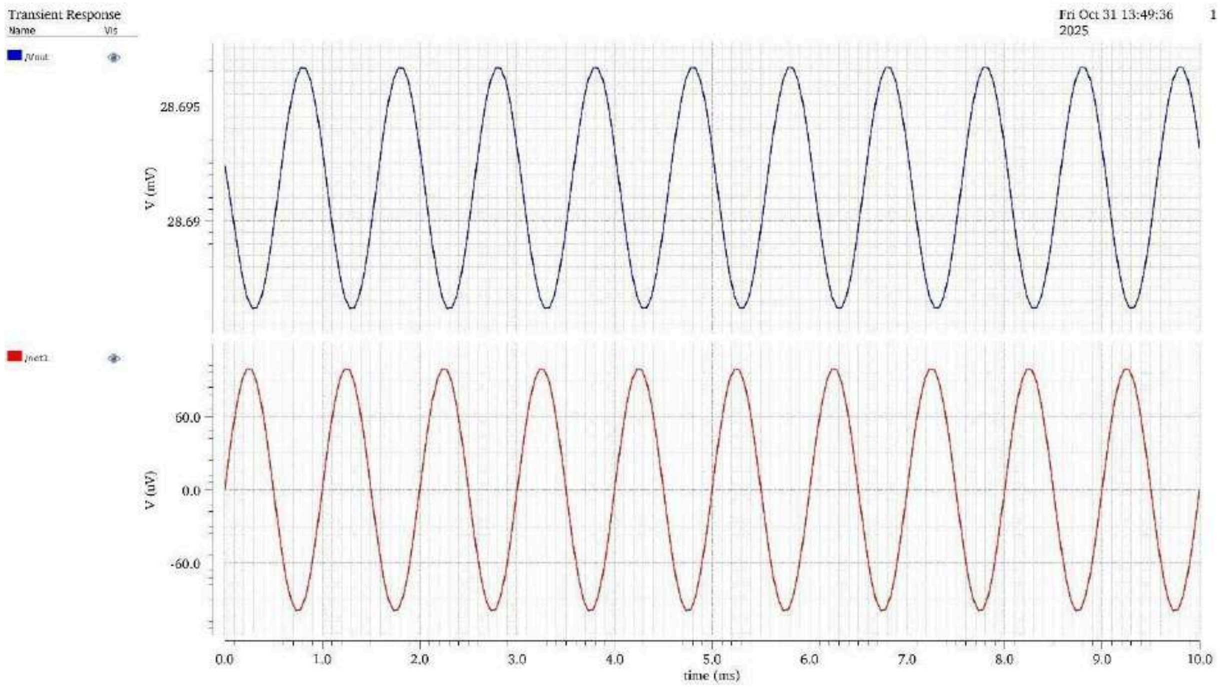

Fig 3: Transient Response of Single Stage Op-Amp

Design Insights

- The achieved gain (~54 dB) is limited by the output resistance of the current mirror load
- Bandwidth is significantly reduced due to the large load capacitance (10 pF)
- Clear tradeoff observed between gain and bandwidth, typical for single-stage amplifiers
- Low bias current successfully reduced power, but also limited transconductance (gm)
- Suitable for low-power applications, but not ideal for high-speed designs

Limitations

- Limited gain compared to multi-stage amplifiers
- Poor bandwidth due to high output node capacitance
- No frequency compensation → not scalable for higher loads

Summary

A compact and power-efficient single-stage CMOS op-amp was designed and verified, demonstrating fundamental analog design tradeoffs between gain, bandwidth, and power.

Two-Stage CMOS Op-Amp (High-Gain Design with Miller Compensation)

Overview

Designed and simulated a two-stage CMOS operational amplifier in Cadence Virtuoso to achieve higher gain and improved output swing compared to a single-stage design, with stability ensured using Miller compensation.

Design Targets

- Gain ≥ 60 dB
- GBW ≈ 30 MHz
- Stable operation with adequate phase margin
- Supply: ±0.9 V
- Load Capacitance: 2 pF
- Slew Rate ≥ 20 V/µs

Architecture

The amplifier consists of two cascaded stages:

- **Stage 1:** NMOS differential amplifier with PMOS current mirror load
- **Stage 2:** Common-source amplifier for additional gain
- **Compensation:** Miller capacitor connected between first and second stage

This topology enables **high overall gain** while maintaining stability through compensation.

Key Design Decisions

- Used **two-stage architecture** to overcome gain limitations of single-stage design
- Implemented **Miller compensation capacitor** to control pole splitting and ensure stability
- Selected **compensation capacitor (~800 fF)** based on load and GBW requirements
- Bias current increased (~8.8 µA) to improve transconductance and bandwidth
- Optimized transistor sizing to balance **gain, speed, and power**
- Ensured proper **pole separation** for stable frequency response

Schematic

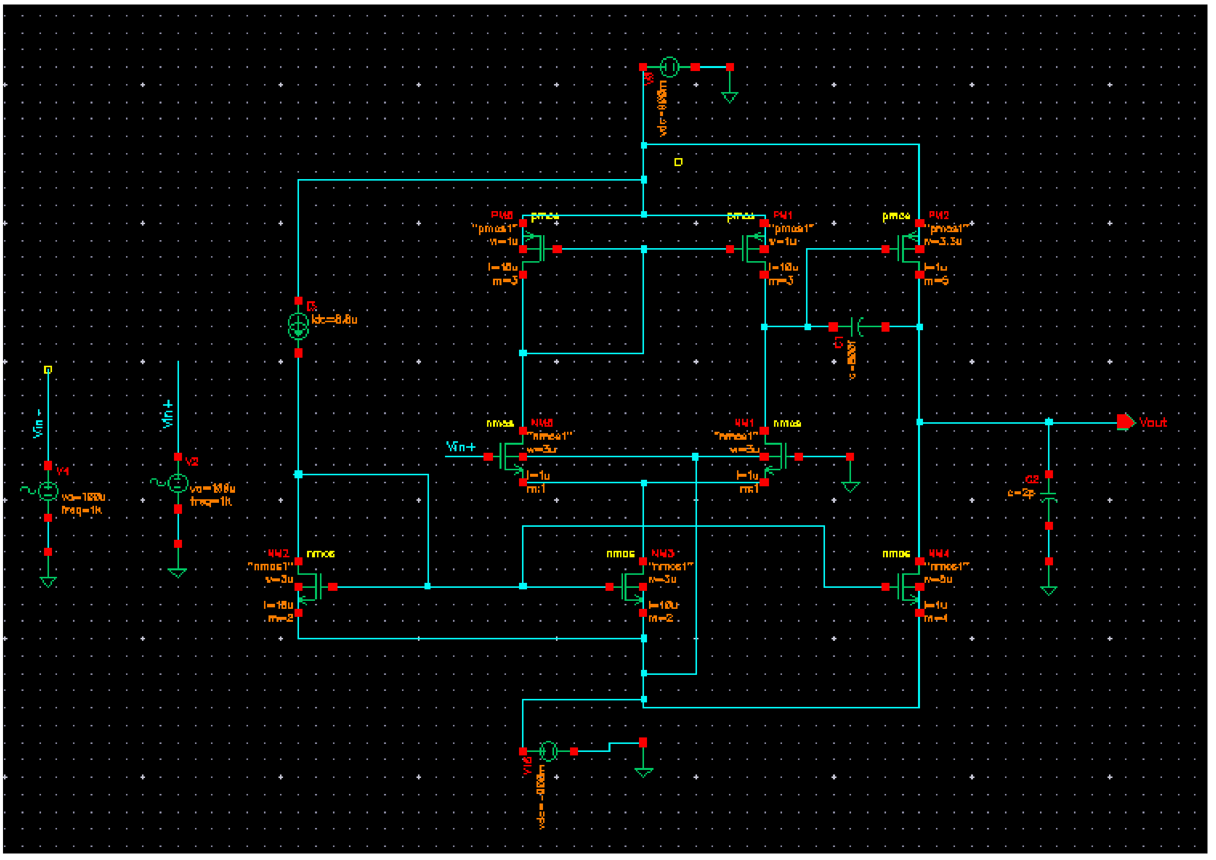

Fig 1: Schematic of Two Stage Op-Amp

Results

Parameter	Value
Gain	58.89 dB
GBW	30 MHz
Bandwidth (-3 dB)	11.76 kHz
Power Consumption	387.31 µW
Slew Rate	20 V/µs
Input Noise	6.02×10^{-10} V²/Hz

Key Plots

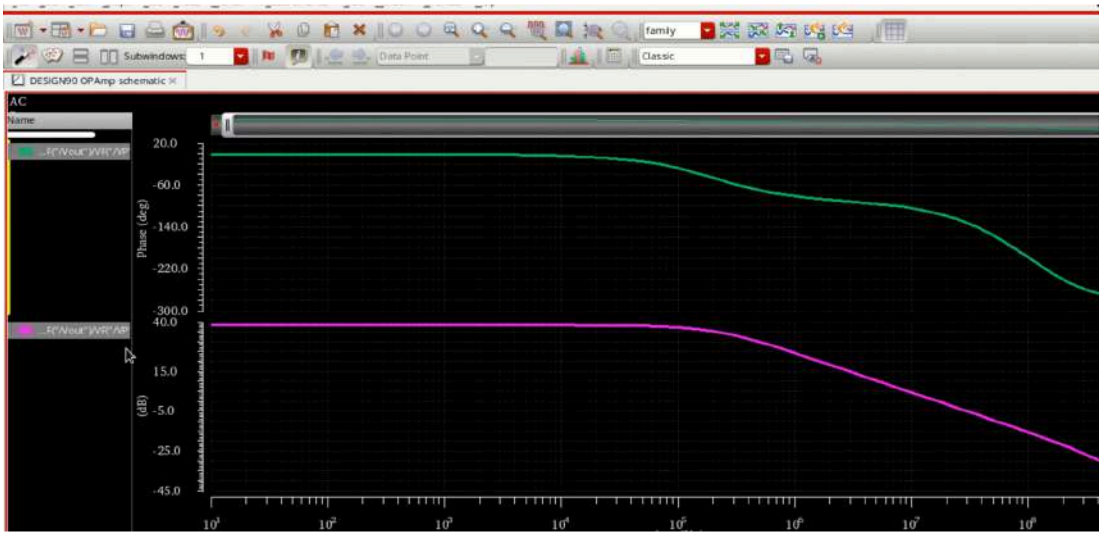

Fig 2: Gain and Phase Plot of Two Stage Op-Amp

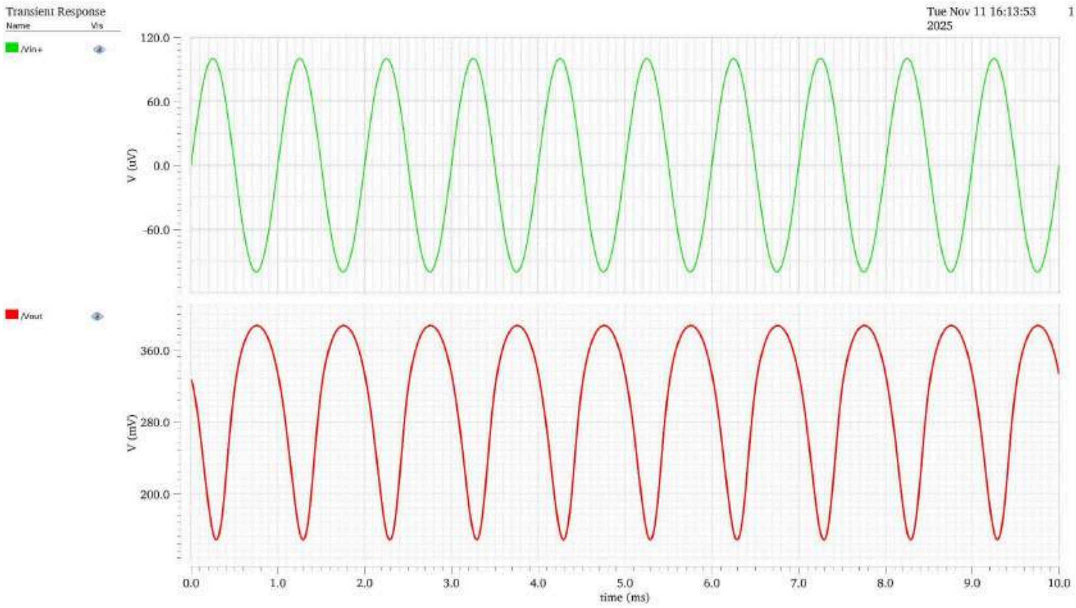

Fig 3: Transient Response of Two Stage Op-Amp

Design Insights

- Gain improvement achieved through **cascading two amplification stages**
- Miller compensation successfully introduced **dominant pole**, ensuring stable operation
- Significant increase in bandwidth compared to single-stage design
- Higher bias current improved **gm**, but increased power consumption
- Tradeoff observed between **gain, bandwidth, and power**

Limitations

- Increased power consumption compared to single-stage amplifier
- Requires careful compensation design for stability
- More complex design and sizing process

Summary

A two-stage CMOS op-amp with Miller compensation was designed to achieve higher gain and bandwidth, demonstrating stability techniques and multi-stage amplifier design tradeoffs.

Bandgap Reference (BGR) – Temperature Independent Voltage Generator (~1.17 V Output)

Overview

Designed and simulated a CMOS Bandgap Reference in Cadence Virtuoso to generate a stable, temperature-independent reference voltage by combining PTAT and CTAT components.

Design Targets

- Output Voltage ≈ 1.2 V
- Temperature Range: -50°C to 125°C
- Near-zero temperature coefficient
- Low power consumption (< 10 µW)
- Supply: 3.3 V

Architecture

The bandgap reference is based on the principle of combining:

- **CTAT voltage** (V_BE of BJT)
- **PTAT voltage** (ΔV_BE across BJTs operating at different current densities)

These components are scaled and summed to produce a **temperature-stable reference voltage**.

Key blocks include:

- Bipolar junction transistors (for V_BE generation)
- Current mirrors
- Resistor network for PTAT scaling
- Biasing circuitry

Key Design Decisions

- Used **PTAT + CTAT compensation** to cancel temperature dependence
- Selected resistor ratios (R1, R2) to accurately scale ΔV_BE
- Designed current mirrors to maintain **stable bias currents**
- Ensured all devices operate in proper regions for consistent behavior
- Targeted **low current operation (~µA range)** to minimize power

Schematic

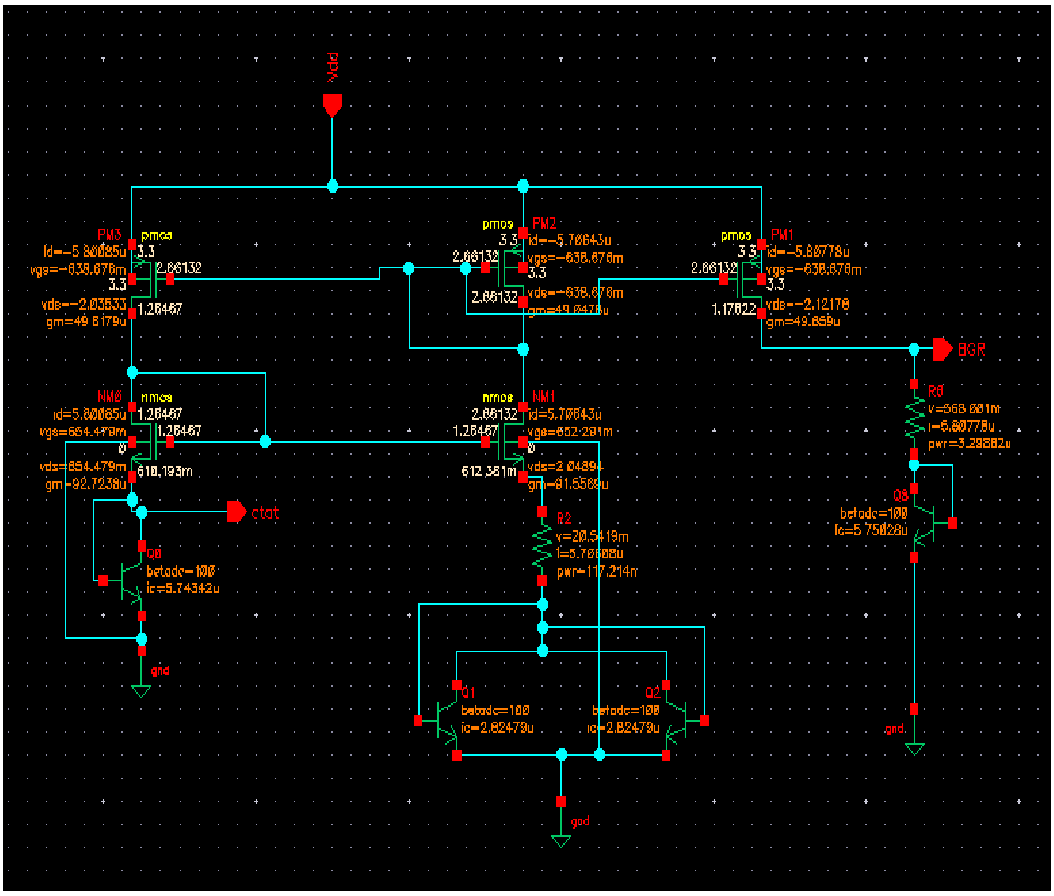

Fig 1: Schematic of Band Gap Reference

Results

Parameter	Value
Output Voltage	~1.17 V
Power Consumption	5.74 µW
Temperature Range	-50°C to 125°C
PTAT Slope	+0.95 mV/°C
CTAT Slope	-1.78 mV/°C
Net Temperature Coefficient	Near 0 mV/°C

Key Plots

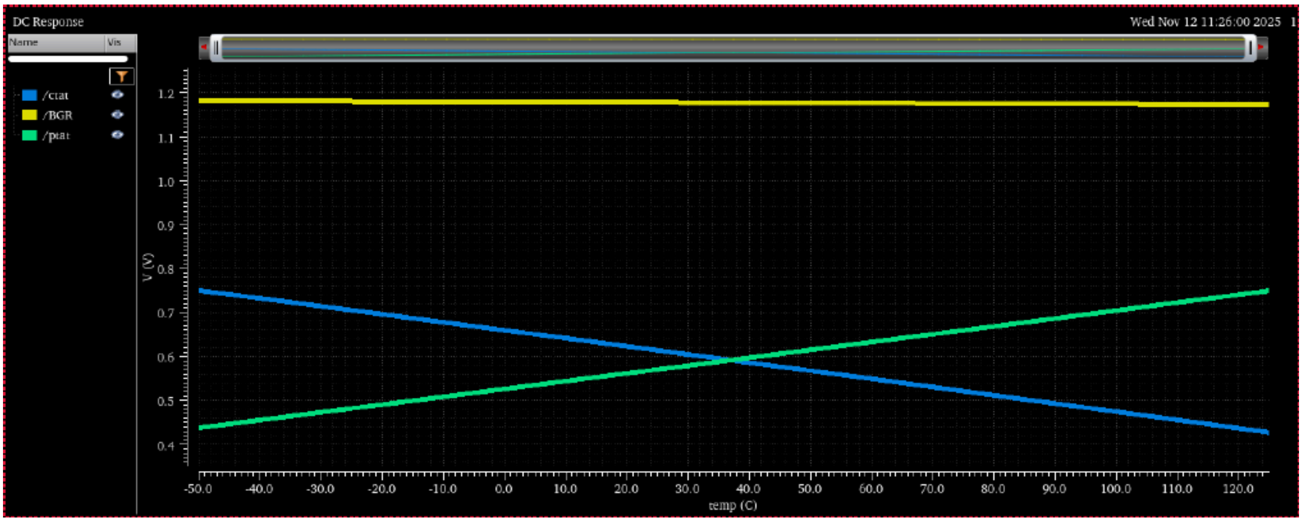

Fig 2: DC Characteristics of PTAT, CTAT, and Bandgap Reference Output

Design Insights

- Successful cancellation of temperature effects achieved by **balancing PTAT and CTAT slopes**
- Output stabilized around **1.17 V**, close to theoretical silicon bandgap reference
- Proper resistor scaling is critical to achieving **zero temperature coefficient**
- Low current design ensures **power-efficient operation**
- Circuit performance is sensitive to **process variations and matching**

Limitations

- Requires precise device matching (especially BJTs and resistors)
- Sensitive to process variations in practical fabrication
- Startup circuit not included (may affect real-world implementation)

Summary

A CMOS bandgap reference was successfully designed to generate a stable ~1.17 V output across a wide temperature range, demonstrating core principles of temperature compensation and precision analog design.

Operational Transconductance Amplifier (OTA) – Low-Power Differential gm Stage

Overview

Designed and simulated a CMOS Operational Transconductance Amplifier (OTA) in Cadence Virtuoso to convert differential input voltage into output current, optimized for low-power analog signal processing applications.

Design Targets

- Moderate voltage gain (~40 dB)
- Low power consumption (< 10 µW)
- High transconductance efficiency (gm/Id)
- Supply: ±0.9 V
- Load Capacitance: 10 pF

Architecture

The OTA is implemented using a single-stage differential structure:

- NMOS differential input pair
- PMOS current mirror active load
- Biasing current source
- Output taken as current (transconductance operation)

Unlike voltage amplifiers, the OTA produces an output current proportional to input differential voltage:

$$I_{out} = gm \times (Vin(+ve) - Vin(-ve))$$

Key Design Decisions

- Designed for high gm/Id to maximize transconductance under low current
- Bias current set to ~8 µA to balance gm and power
- Used current mirror load to achieve high output resistance
- Ensured all devices operate in saturation region
- Optimized sizing to improve linearity and transconductance

Schematic

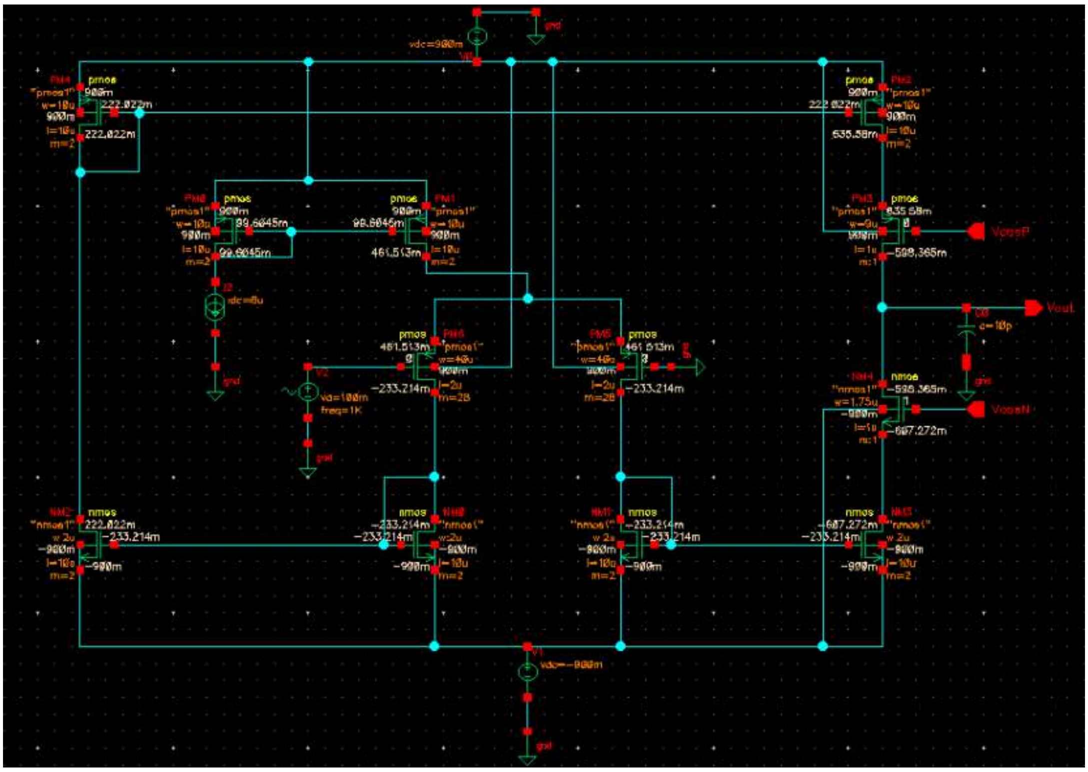

Fig 1: Schematic of Operational Transconductance Amplifier

Results

Parameter	Value
Gain	39.91 dB
Transconductance (gm)	~100 µS (input pair)
Bandwidth (-3 dB)	25.32 kHz
Power Consumption	7.39 µW
Input Noise (RMS)	4.60×10^{-12} V²

Key Plots

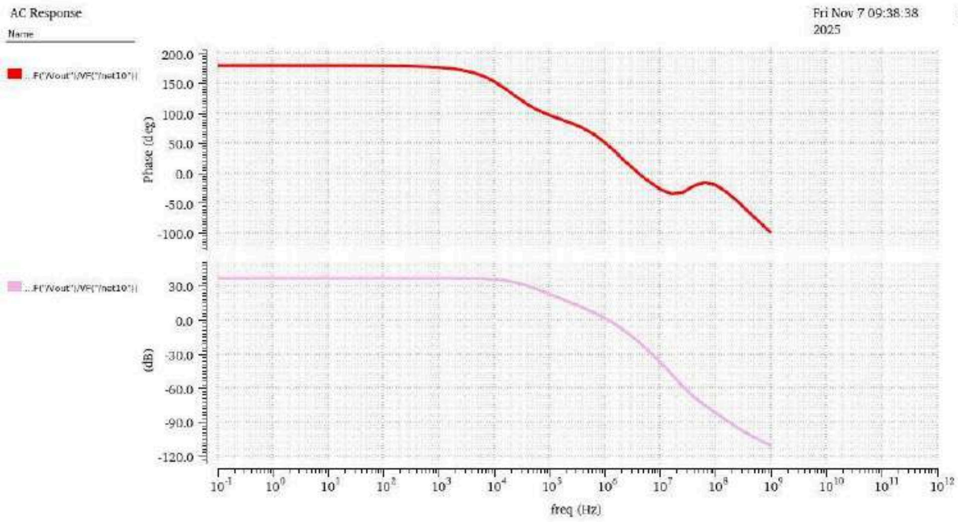

Fig 2: Gain and Phase Plot of Operational Transconductance Amplifier

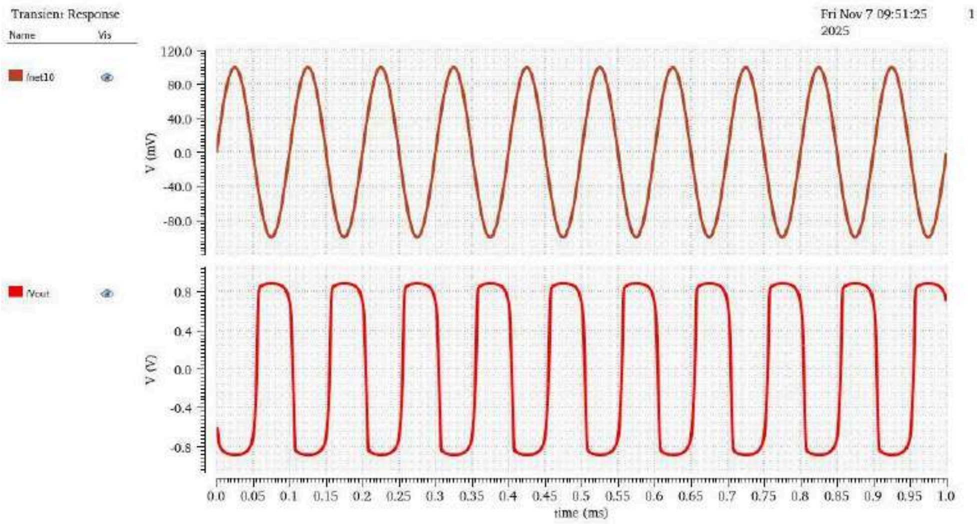

Fig 3: Transient Response of Operational Transconductance Amplifier

Design Insights

- OTA performance is primarily governed by transconductance (gm) rather than voltage gain.
- Higher gm is achieved by increasing bias current, but at the cost of power.
- Output current behavior makes OTA suitable for Gm-C filters and analog signal processing.
- Gain (~40 dB) is lower than op-amps due to single-stage architecture.

Limitations

- Limited voltage gain compared to multi-stage amplifiers
- Output is current-based → requires load or conversion for voltage output
- Sensitive to bias current variations

Summary

A low power CMOS OTA was designed to achieve efficient voltage to current conversion that demonstrates key tradeoffs between transconductance, gain, and power in analog design.